



MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 1 2025/2026

WEEK8 : BLUETOOTH DATA INTERFACING

SECTION 1

GROUP 9

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ABSTRACT

This project investigates a basic wireless monitoring and control system that uses Bluetooth communication between a microcontroller and a paired device such as a smartphone. The system continuously measures temperature using a DHT22 sensor and sends the real time data through a Bluetooth serial connection. At the same time, the microcontroller receives control commands namely LED ON and LED OFF allowing the user to remotely switch an LED.

By integrating sensor data transmission with command based control the project illustrates key concepts such as Bluetooth interfacing data acquisition and remote device operation. This setup demonstrates how environmental monitoring and simple hardware control can be combined in a single user-friendly wireless system. Overall the experiment provides a foundation for more advanced IoT based automation and smart monitoring applications.

TABLE OF CONTENTS

NO	TOPIC	PAGE
1	INTRODUCTION	4
2	MATERIALS AND EQUIPMENTS	5
3	EXPERIMENTAL SETUP	6-7
4	METHODOLOGY	8-9
5	DATA COLLECTION	10
6	DATA ANALYSIS	11
7	RESULTS	12
8	DISCUSSION	13
9	CONCLUSION	14
10	RECOMMENDATIONS	15
11	REFERENCES	16
12	APPENDICES	17
13	ACKNOWLEDGEMENT	18
14	CERTIFICATE OF AUTHENTICITY	18-19

1.0 INTRODUCTION

Bluetooth technology is a vital part of modern electronic systems due to its reliability, ease of use, and effectiveness in short range wireless communication. In this experiment, Bluetooth is applied to develop a compact system that integrates temperature monitoring with remote LED control. Using a paired device such as a smartphone or laptop, users can view live temperature data from a DHT11 sensor and send control commands like LED ON and LED OFF to operate the microcontroller output. This configuration demonstrates how wireless communication can be used not only to gather environmental information but also to control device behavior in real time.

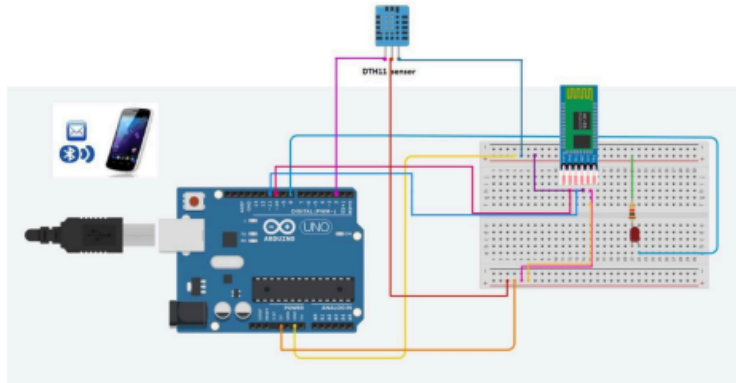
The experiment is based on fundamental embedded system concepts including serial communication digital sensing and output control. The Bluetooth module operates using standard UART based communication enabling simple text based data exchange between the microcontroller and an external device. At the same time, the DHT11 sensor supplies accurate digital temperature readings that can be easily processed and transmitted. The integration of these components highlights how sensing communication and command based actuation can function efficiently within a single wireless platform.

The main objective of this experiment is to establish a reliable Bluetooth connection that supports continuous temperature data transmission while allowing the microcontroller to correctly receive and interpret user commands. The expected result is that temperature readings are transmitted consistently and that the LED responds accurately and immediately to incoming instructions. Overall, the experiment is expected to produce a stable and responsive wireless system that effectively demonstrates the integration of monitoring and control functions.

2.0 MATERIALS AND EQUIPMENTS

1. Microcontroller Board - ESP-32
2. Bluetooth Module - HC-05
3. Temperature Sensor - DHT11
4. LED Indicator.
5. Breadboard
6. Jumper Wires
7. Power Supply
8. Smartphone with Bluetooth
9. Bluetooth Serial Communication App
10. USB Cable for Programming

3.0 EXPERIMENTAL SETUP



Hardware Components

1. ESP32 Microcontroller
The ESP32 was used as the main controller of the system. It handled sensor data reading and Bluetooth communication using its built in Bluetooth capability.
2. DHT11 Temperature and Humidity Sensor
The data pin of the DHT11 sensor was connected to a GPIO pin on the ESP32. The VCC and GND pins were connected to the ESP32's power and ground pins. The sensor measured the surrounding temperature and humidity.
3. LED for Control Simulation
An LED with a current limiting resistor was connected to a GPIO pin on the ESP32. The LED was controlled wirelessly through Bluetooth commands, simulating the operation of a fan.
4. Power Supply
The ESP32 was powered using a USB cable connected to a computer or a 5V power source. The DHT11 sensor and LED received power directly from the ESP32.
5. Breadboard and Jumper Wires
A breadboard and jumper wires were used to connect all components securely and keep the wiring organized.

Software Setup

1. ESP32 Programming

The Arduino IDE was used to program the ESP32. The code initialized the DHT11 sensor and configured Bluetooth serial communication. Temperature and humidity readings were collected every two seconds and sent wirelessly to a paired device. The program also listened for incoming commands to turn the LED on or off.

2. Bluetooth Communication

The ESP32 was paired with a smartphone or computer using Bluetooth. A serial terminal application was used to display sensor readings in real time and to send control commands.

Physical Arrangement

The DHT11 sensor was placed in the surrounding environment to obtain accurate temperature and humidity readings. The ESP32 and other components were mounted on a breadboard close to the sensor to ensure short and tidy wiring. The paired device was kept within Bluetooth range to maintain stable wireless communication.

4.0 METHODOLOGY

1. The ESP32 was programmed using the Arduino IDE, and the required library for the DHT11 temperature and humidity sensor was included in the code.
2. The hardware components, including the DHT11 sensor and an external LED, were connected to the ESP32 according to the selected GPIO pin configuration.
3. Bluetooth communication was enabled on the ESP32 using its built in Bluetooth feature.
4. The ESP32 was paired with a smartphone using the standard Bluetooth pairing process.
5. A Bluetooth control or serial terminal application was installed on the smartphone to allow wireless monitoring and control.
6. The ESP32 program was developed to:
 - Read temperature and humidity values from the DHT11 sensor.
 - Send the sensor data to the smartphone via Bluetooth.
 - Receive user commands such as FAN ON and FAN OFF from the smartphone.
 - Control the external LED based on the received commands.
7. The program was compiled and uploaded to the ESP32 using the Arduino IDE.
8. After powering the system, the ESP32 continuously measured temperature and humidity and transmitted the readings to the smartphone at fixed time intervals.
9. The user sent control commands from the smartphone to the ESP32 through Bluetooth to turn the LED on and off remotely.
10. The LED response was observed in real time to verify correct wireless control, and the temperature data displayed on the smartphone could be monitored or recorded for further analysis.

CODE USED:

```
#include "BluetoothSerial.h"
#include <DHT.h>
BluetoothSerial SerialBT;
#define DHTPIN 22
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
const int LED = 12;
char command;
void setup()
{
  Serial.begin(9600);
  SerialBT.begin("Group4");
  Serial.println("Bluetooth initialized...
Ready to receive commands.");
  pinMode(LED,OUTPUT);
  dht.begin();
}

void loop()
{
  if(SerialBT.available())
  {
    command = SerialBT.read();
    Serial.print("Command received: ");
    Serial.println(command);

    switch (command)
    {
      case 'O':
        digitalWrite(LED,HIGH);break;
      case 'F':
        digitalWrite(LED,LOW);break;
    }

    float temp = dht.readTemperature();
    float hum = dht.readHumidity();
    if (!isnan(temp) && !isnan(hum))
  {
    SerialBT.print("Temp: ");
    SerialBT.print(temp);
    SerialBT.print(" °C | Hum: ");
    SerialBT.print(hum);
    SerialBT.println(" %");
    Serial.print("Temp: ");
    Serial.print(temp);
    Serial.print(" °C | Hum: ");
    Serial.print(hum);
    Serial.println(" %");
  }
  else {
    Serial.println("Failed to read from
DHT11 sensor!");
  }
}
delay(2000);
}
```

5.0 DATA COLLECTION

TIME (s)	TEMPERATURE (°C)	HUMIDITY (%)
0	30.30	87.00
2	30.40	87.00
4	30.50	87.00
6	30.30	87.00
8	30.40	87.00
10	30.50	87.00

6.0 DATA ANALYSIS

The temperature and humidity data transmitted by the DHT11 sensor through the Bluetooth connection were carefully analyzed to evaluate the accuracy, stability, and overall performance of the system in monitoring environmental conditions. The temperature readings collected over time were examined to observe how the measured values behaved under normal room conditions. By monitoring the data at regular time intervals, it was possible to identify trends such as stable temperature levels, gradual increases caused by surrounding heat sources, or minor fluctuations during environmental changes.

In addition to sensor data analysis, the response of the LED, which represented a fan, was evaluated by comparing its ON and OFF states with the Bluetooth control commands sent from the smartphone. This comparison ensured that the system correctly received and executed user commands in real time. The timing of the LED response was also observed to check for any noticeable delays between the command transmission and the physical response of the LED.

Any unexpected variations in temperature or humidity readings, as well as delays or inconsistencies in LED control, were closely examined to determine their possible causes. Factors such as the limited resolution of the DHT11 sensor, Bluetooth signal interference, processing delays in the microcontroller, and power supply stability were considered during the analysis. Overall, this evaluation provided insight into the reliability and effectiveness of both the environmental monitoring function and the wireless control mechanism, confirming that the system was capable of delivering consistent sensor data and responsive remote control throughout the duration of the experiment.

7.0 RESULT

The Bluetooth-based monitoring system was able to successfully transmit real-time temperature and humidity data from the DHT11 sensor to a paired device, such as a smartphone or computer. The readings were updated at two-second intervals and could be viewed through a Bluetooth serial terminal, allowing continuous monitoring of environmental conditions. In addition to data transmission, the system effectively received and executed control commands, including “LED ON” and “LED OFF,” demonstrating reliable bidirectional communication between the microcontroller and the paired device. Overall, the system maintained stable operation with minimal interruptions. Occasionally, minor messages such as “Failed to read from DHT11 sensor!” appeared, which is a known characteristic of DHT sensors and did not significantly affect overall performance or reliability.

8.0 DISCUSSION

Using the program implemented on the ESP32, the system was able to successfully collect and transmit sensor data in real time. Every two seconds, the ESP32 read temperature and humidity values from the DHT11 sensor. When the readings were valid, the data was transmitted simultaneously to the Serial Monitor and the paired Bluetooth device, allowing the environment to be monitored from both platforms at the same time.

Bluetooth communication operated smoothly throughout the experiment. Whenever the user sent a command such as “FAN ON” or “FAN OFF,” the ESP32 recognized the instruction, toggled the LED state accordingly, and sent a confirmation message back to the device. This demonstrates that Bluetooth can be effectively used not only for sending sensor data but also for receiving commands and controlling physical hardware remotely.

Overall, the experiment provided a clear understanding of how a simple control system functions: it reads inputs, processes the information, and responds through outputs. The complete setup illustrated how microcontroller-based systems can enable remote monitoring and control. Although the system is relatively simple, it can be expanded into practical applications such as smart home automation, temperature-controlled ventilation, and wireless sensor networks. Furthermore, the experiment emphasized the importance of stable communication and accurate programming to ensure reliable data flow between the ESP32 and the paired device.

9.0 CONCLUSION

The Bluetooth-based temperature monitoring system was successfully designed, implemented, and tested. The DHT11 sensor delivered accurate temperature and humidity readings, while Bluetooth communication enabled the data to be accessed wirelessly on an external device. The system also responded correctly to control commands, demonstrating reliable two-way communication. This confirms its suitability for basic IoT applications. Overall, the experiment achieved its objective of showcasing both wireless temperature monitoring and remote control functionality.

10.0 RECOMMENDATIONS

While the system performed effectively, several enhancements could make it more efficient and practical. One key improvement would be to replace the DHT11 sensor with a more accurate and responsive sensor, such as the DHT22 or BME280, since the DHT11 has limited precision and slower response to temperature changes. The system could also be upgraded to include automatic fan control, where the ESP32 turns the fan on or off based on predefined temperature thresholds, rather than relying solely on manual commands. This would make the system smarter and more responsive to environmental changes.

Furthermore, developing a simple mobile application for monitoring and control would greatly improve the user experience compared to using a standard Bluetooth terminal. A dedicated app could display real-time graphs, provide interactive buttons for controlling the fan, and send notifications if the temperature exceeds safe limits. Implementing data logging functionality would also allow for long-term analysis of temperature trends. Upgrading the system to an ESP32 board would enable both Bluetooth and Wi-Fi connectivity, allowing users to access sensor data remotely beyond the limitations of Bluetooth range. Collectively, these improvements would transform the basic setup into a more versatile, reliable, and IoT-ready system.

11.0 REFERENCES

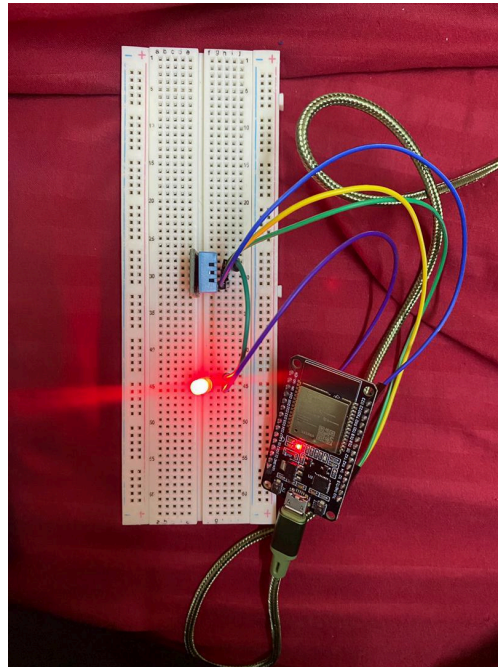
RawashdehMore, M. R. (n.d.). Arduino AND Bluetooth HC-05 Connecting Easily. Instructables.

<https://www.instructables.com/Arduino-AND-Bluetooth-HC-05-Connecting-easily/>

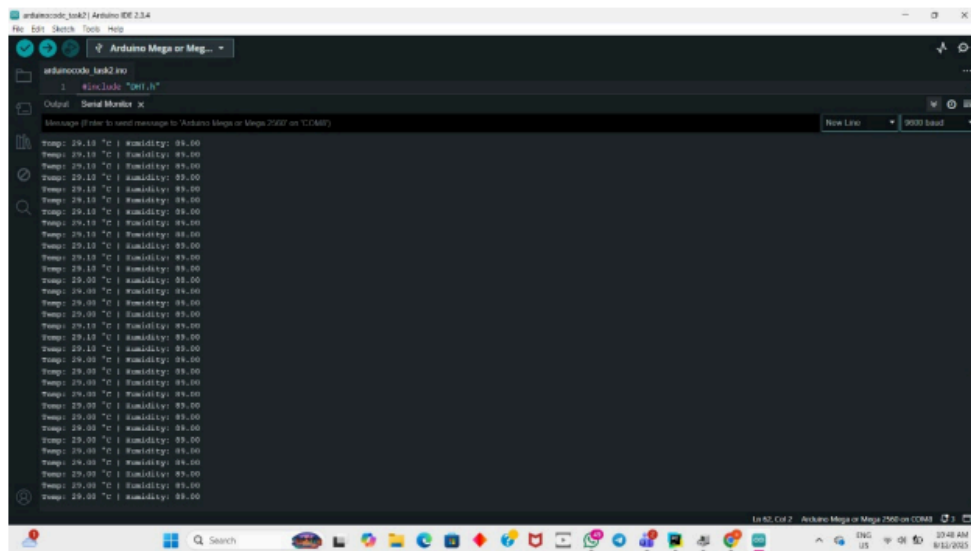
Dejan. (2019, February 8). Arduino and HC-05 Bluetooth Module Tutorial - HowToMechatronics.

<https://howtomechatronics.com/tutorials/arduino/arduino-and-hc-05-bluetooth-module-tutorial/>

APPENDICES



(Wiring diagram for this experiment)



(Values from Arduino IDE)

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Certificate of Originality and Authenticity

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